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New paleomagnetic results for Ordovician and Silurian rocks of the Tarim Block, Northwest China and their paleogeographic implications

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ABSTRACT

The kinematics of the Tarim Block is poorly understood prior to the late Paleozoic and this study aims to clarify motions during Ordovician and Silurian times. We report the paleomagnetic record from 12 sites in Ordovician marine sediments and 21 sites in Silurian fine-grained sandstones and mudstones from the Aksu and Kalpin areas of northwestern Tarim. Together with our previously reported preliminary results, the Ordovician sediments isolate a sporadically-preserved pre-Cenozoic characteristic remanence with dual polarity. Although a positive fold test can only define the remanence as pre-Cenozoic, the corresponding paleomagnetic pole (33.7°S, 185.0°E, dp/dm = 2.7°/4.0°) is removed from post-Silurian paleopoles from Tarim and combined with evidence from magnetic fabrics indicating a quasi-primary origin. A characteristic remanence isolated at 17 sites in Silurian sediments from the Aksu and Kalpin areas includes 7 sites with suspected contamination from regional Permian igneous activity. The 10 remaining sites, all from Kalpin, yield a positive fold test and a paleopole at 19.1°N, 172.9°E (A_{95} = 5.5°) significantly removed from younger paleopoles from Tarim indicating a Late Llandovery to Wenlock remanence age. The new Mid-Late Ordovician paleopole places the Tarim Block at ~32°S, probably in the vicinity of the Arabian margin of Gondwana. Rapid APW between ~460 and ~433 Ma records large differential dextral motions of Tarim together with other continental blocks along the Paleo-Asian Ocean margin which brought the China continental blocks into proximity with Australia by mid-Silurian times in low (~9°N) northerly paleolatitudes.

1. Introduction

The Tarim Basin of central Asia is located between the Kunlun, Tianshan and Altyn Mountain ranges. It is the largest inland basin within China, covering an area of ~530,000 km^2. The basin, rooted in Archean crust and together with the bordering mountain ranges, forms one of three major continental blocks of Eastern Asia. Resolving the Paleozoic tectonic evolution of the Tarim Block is an important facet of the transfer history of the Eastern Asian blocks from the East Gondwana margin to the northern branch of the Pangean Supercontinent, and thus to the evolution of the Paleo-Asian Ocean (PAO) tectonic domain. Paleozoic paleomagnetic studies in the North and South China blocks, particularly during the last decade of the 20th century (e.g., Opdyke et al., 1987; Steiner et al., 1989; Enkin et al., 1992; Zhao et al., 1992, 1993; Embleton et al., 1996; Yang et al., 1998, 2002; Bai et al., 1998; Huang et al., 1999, 2000, 2001; Han et al., 2015), have provided reliable constraints on their Paleozoic paleogeography and kinematics (e.g., Enkin et al., 1992; Yang et al., 1998, 2004; Zhao et al., 1996; Cocks and Torsvik, 2013; Huang et al., 2018). In contrast Paleozoic paleomagnetic results from the Tarim Block meeting modern reliability criteria (e.g., Van der Voo, 1990) have remained scarce. Specifically the early Paleozoic interval is devoid of results with the exception of a preliminary Early Ordovician paleopole from three dolomitic limestone sites in the Kuruktage area of northeastern Tarim reported by Fang et al. (1996). Specific challenges to paleomagnetic studies in this region include (i) a strong late Paleozoic crust-mantle interaction across the PAO tectonic domain which resulted in large-scale magmatism and consequent regional overprinting (Natal'in and Sengör, 2005; Xiao et al., 2009); (ii) complex Cenozoic tectonic deformation resulting in crustal shortening and block extrusion in central Asia induced by the India-Asia collision (e.g., Molnar and Tapponnier, 1975; Le Pichon et al., 1992; Avouac et al., 1993; Yin and Harrison, 2000; Tapponnier et al., 2001; Johnson, 2002); (iii) widespread regional blanketing by the Taklimakan Desert which has left only sporadic exposures of early

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Table 1
Summary of Ordovician paleomagnetic results for characteristic B component from the Aksu-Kalpin areas of northwestern Tarim.

Site ID	Lithology	Location		Bedding	n/n_0	R/N	In-situ		Tilt-corrected		k	α_{95}(°)	VGP[c]	
		φ_s(°E)	λ_s(°N)	Strike/dip			D_g(°)	I_g(°)	D_s(°)	I_s(°)			φ_p(°E)	λ_p(°N)
as20[a]	Sandy limestone	79.8345	40.8396	79/61	3/10	0/3	140.1	-6.8	112.5	-54.9	54.6	16.8	187.1	-38.0
as22[a]	Sandy limestone	79.8345	40.8396	81/66	4/10	0/4	137.8	1.8	115.1	-48.7	61.1	11.8	179.4	-37.0
ks89 + 90[a]	Marl, mudstone	78.9203	40.5585	234.5/41	10(8)/18	0/10	16.4	-80.2	128.9	-56.0	29.4	9.7	180.2	-50.4
ks91[a]	Sandy limestone	78.9218	40.5587	239/42	10(1)/11	0/11	2.8	-74.9	132.4	-59.6	51.8	6.8	184.7	-54.3
ks92[a]	Marl, mudstone	78.9240	40.5586	240/39	8(2)/9	0/8	56.9	-53.0	98.4	-39.9	216.8	3.8	181.6	-20.7
xk205	Nodular limestone	78.9259	40.5583	242.6/37	9/9	9/0	222.0	44.9	261.4	45.6	230.1	3.4	15.1	11.2
xk206	Nodular limestone	78.9223	40.5549	242.6/37	9(2)/9	9/0	203.7	55.2	267.7	60.6	77.1	5.9	23.9	24.1
xk207[b]	Nodular limestone	78.9219	40.5550	238.6/34	8(1)/9	0/8	330.0	33.2	330.0	-0.8	47.7	8.1	300.3	40.8
xk208	Nodular limestone	78.9243	40.5585	235.6/37	5(5)/9	0/16	66.9	-68.8	115.0	-46.0	35.2	6.3	176.4	-35.7
xk209	Marl	78.9244	40.5586	237.6/34	5(5)/9									
xk210	Marl	78.9212	40.5551	235.6/36	6(3)/9									
xk211	Limestone, marl	78.9212	40.5551	238.6/37	1(1)/8	1/10	122.4	-72.2	138.7	-37.4	20.2	10.4	152.2	-49.9
xk212	Limestone, marl	78.9210	40.5552	238.6/36	9(9)/9									
xk213	Limestone, marl	78.9208	40.5552	237.6/37	1/9									
xk214	Limestone, marl	78.9207	40.5552	237.6/37	7(7)/8	0/11	91.8	-78.7	134.6	-44.4	75.2	5.3	162.2	-49.9
xk215	Limestone, marl	78.9205	40.5551	236.6/39	2(2)/9									
xk216	Limestone, marl	78.9202	40.5550	235.6/41	2(1)/8									
xk217	Limestone, marl	78.9198	40.5552	231.6/37	9(2)/9	0/9	28.0	-75.6	117.4	-56.5	46.8	7.6	186.1	-42.1
Formation Mean		by sites			11/12	2/9	82.0	-66.8			4.8	23.3	182.2	-38.4
									115.1	-51.4	31.3	8.3	$K = 21.3$, $A_{95} = 10.1$	
		by specimens			100(48)/172		56.7	-67.3			7.3	5.5	185.0	-33.7
									108.7	-51.8	23.9	2.9	$dp/dm = 2.7/4.0$	

Fold test:
(1) Watson and Enkin's test (1993): the optimum concentration (maximum k) is achieved at 99.7 ± 6.8% unfolding;
(2) McFadden's test (1990): $N = 11$, in-situ $\xi_2 = 7.977$, tilt-corrected $\xi_1 = 0.320$, statistical threshold $\xi = 3.865$ at 95% confidence level, indicative of a positive fold test.
Abbreviations are: site ID, site identification; Strike/dip, strike azimuth and dip of bed; n/n_0, number of samples used in mean calculation/demagnetized, number in parentheses indicates number of remagnetization circles used; R/N, samples showing reversed/normal polarity; D_g, I_g (D_s, I_s), declination and inclination of direction in-situ (after tilt-correction); λ_s, φ_s, latitude and longitude of the sampling site; λ_p, φ_p, latitude and longitude of corresponding virtual geomagnetic pole (VGP) to tilt-corrected site-mean directions.
[a] Site-mean directions from Sun and Huang (2009) with that of site as22 being recalculated.
[b] Site-mean direction discarded from final formation mean calculation.
[c] VGP positions corresponding to tilt-corrected site-mean directions.

Paleozoic strata currently confined to the Kalpin and Kuruktage stratigraphic subdivisions (Jia et al., 2004) at the northern margin of the basin where the rocks have further been subjected to deformation by post-late Paleozoic tectonism; and (iv) lithological constraints limited by an absence of igneous rocks suites and a record confined to a sedimentary succession often weathered, and of varying suitability for investigation.

To constrain the early Paleozoic paleogeography of Tarim and thus provide key controls for resolving the kinematic history of this tectonic block we conducted paleomagnetic sampling of Ordovician and Silurian rocks comprising outcrops at the northern margin of the Tarim Block during the first decade of the present century. Preliminary rock magnetic and paleomagnetic results from 432 Ordovician samples occupied at 44 sampling sites were reported in Chinese by Sun and Huang (2009) where it was found that only 35 relatively pure limestone samples ferromagnetically (sensu *lato*) dominated by magnetite yielded a likely primary or quasi-primary remanence component (Table 1). In addition to the removal of the pole from younger results, the possible - primary nature of this remanence was suggested by a significant positive fold test at both sampling site and specimen levels although the history of regional tectonism was only able to use this to define a pre-Cenozoic age. In contrast the majority of the Ordovician collection yielded a post-folding Cenozoic overprint component in limestones (sometimes argillaceous) and marls where hematite and subordinate magnetite were present as magnetic carriers.

With the aim of reinforcing the reliability of the preliminary nature of an Ordovician paleopole based on just 35 samples, we have revisited the Kalpin area and collected new samples from the Mid-Ordovician Saergan Formation which the earlier investigation suggested would be the most promising target. In this paper we report the full experimental results from this new Ordovician collection and a companion Silurian collection, and we develop the paleogeographic interpretation of the results. Combined with existing late Neoproterozoic and Paleozoic paleomagnetic results from Tarim, we show that it is now possible to monitor in outline the tectonic evolution of the Tarim Block within the PAO from early Paleozoic times and thereby contribute to understanding the evolutionary history of this marginal sector of the Eurasia-East Gondwana continental domain.

2. Geological setting and paleomagnetic sampling

Sandwiched within the Tianshan Mountains in the north, the Kunlun Mountains and the Altyn Mountains in the south (Fig. 1a), the Tarim Basin records a large inland Meso-Cenozoic compound basin superimposed onto a Paleozoic platform (e.g., Zhou, 2001). Arid conditions within the vast Taklimakan Desert in the central part of the basin have swept a superficial cover across the basin so that rock outcrops, and specifically early Paleozoic successions, are mainly confined to the Aksu-Kalpin-Bachu (AKB) area in the northwest, designated as the Kalpintage stratigraphic region by Jia et al. (2004), and the Kuruktage area in the northeast; in addition sporadic outcrops occur at the southern slope in the western part of the South Tianshan Mountains. In the AKB area early Paleozoic strata in particular, are well-developed and constrained by the paleontological record, although lower Neoproterozoic, Upper Devonian to Carboniferous, and Mesozoic rocks are absent (Jia et al., 2004).

The Cambrian and Ordovician succession consists mainly of nearshore to shallow marine carbonate facies (Fig. 1) with the Ordovician following the Upper Cambrian (Xiaqiulitage Formation) conformably and succeeded with disconformity by the Lower Silurian Kalpintage

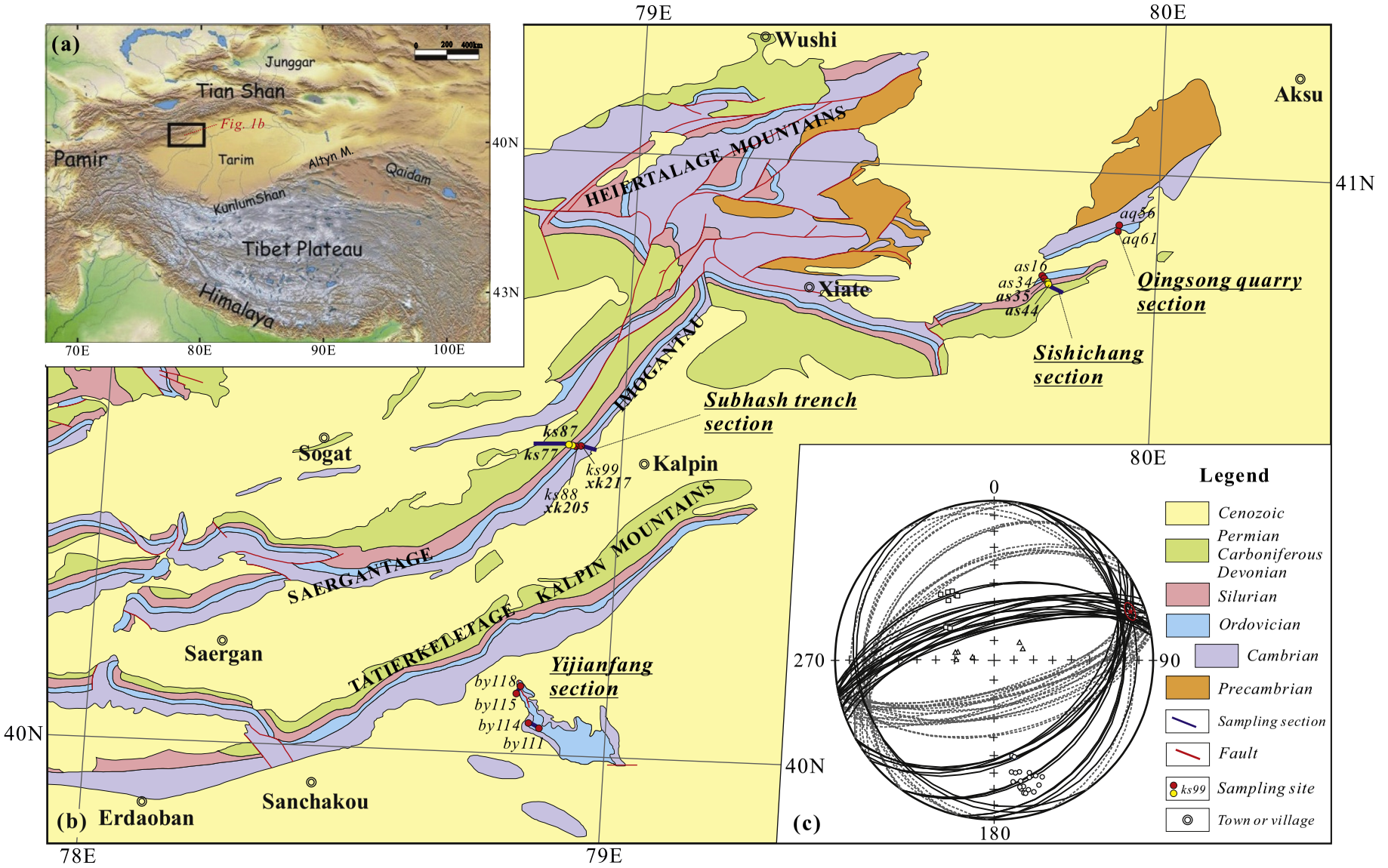


Fig. 1. (a) Topographic map of the Tibetan Plateau and central Asia showing the tectonic framework of the study area and surroundings; (b) simplified geologic map of the Aksu-Kalpin-Bachu (AKB) areas, northwestern Tarim with the distribution of paleomagnetic sampling sections and sites of this study (sampling sites in bold) and Sun and Huang (2009); (c) stereonet of bedding planes from this study on the upper hemisphere highlighting the nearly-horizontal fold axes of deformation influencing Ordovician strata in northwestern Tarim.

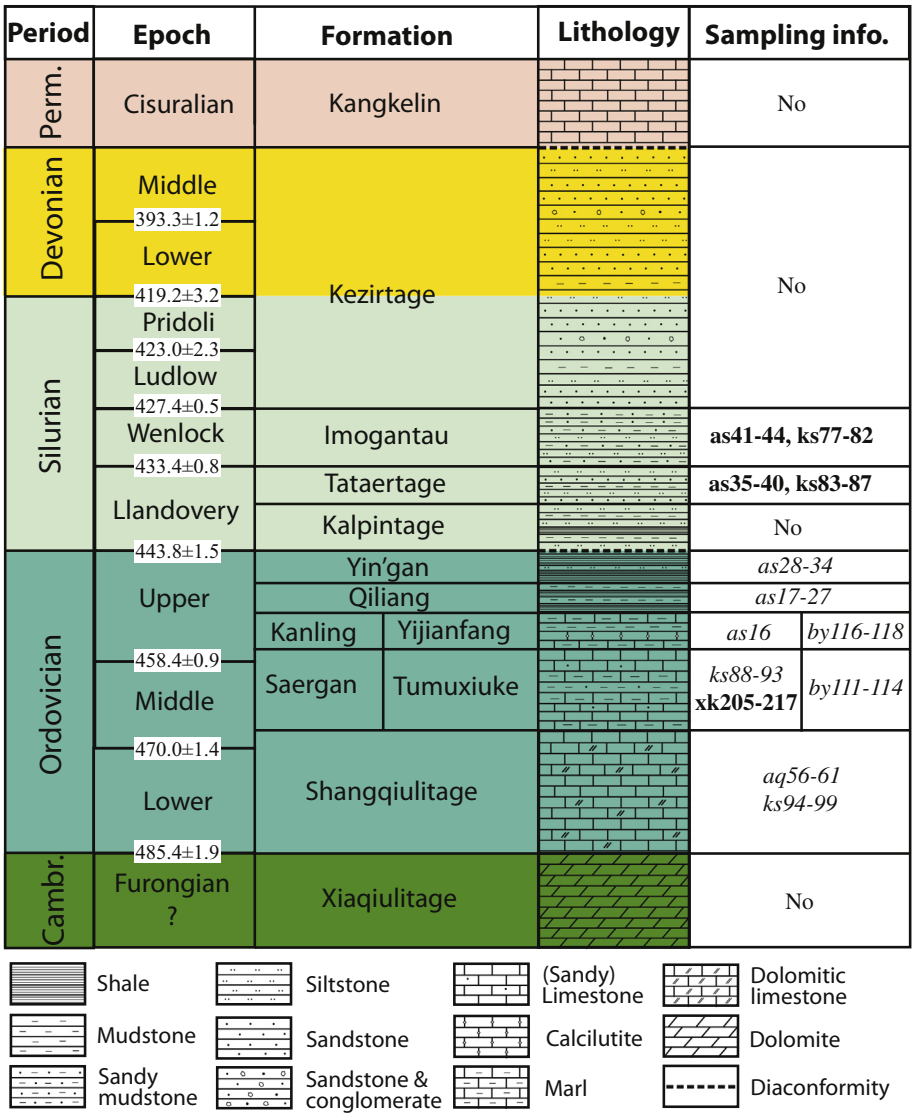


Fig. 2. Schematic lithostratigraphic column illustrating Paleozoic strata in the Aksu-Kalpin-Bachu areas of northwestern Tarim (Jia et al., 2004); sampling sites in bold letters indicate those reported in this study. Numerical ages (Ma) for epoch boundaries are referred to the International Chronostratigraphic Chart v2018/08 (http://www.stratigraphy.org/ICSchart/ChronostratChart2018-08.pdf).

Formation (Fig. 2). The Ordovician Shangqiulitage Formation is widely exposed across the AKB region and comprises mainly limestones, sometimes dolomitic, with an abundance of trilobites and cephalopods. The Saergan Formation is mostly exposed in the Aksu-Kalpin area and consists largely of limestones, marls and mudstones sporadically interbedded with more arenaceous members. An abundant Middle to Late Ordovician fauna including trilobites, brachiopods, gastropods, cephalopods and graptolites in the Dawangou Section northwest of the village of Yin'gan in Kalpin County permits an accurate paleontological age assignment. The Yijianfang and Tumuxiuke formations, although exposed only in the Bachu area, are biostratigraphically comparable to the (Middle to Upper Ordovician) Saergan and (Upper Ordovician) Kanling formations respectively and comprise similar lithologies of limestones and marls containing an abundant fossil record permitting a refinement of the regional paleontological control (Zhou, 2001; Jia et al., 2004). In the Aksu-Kalpin areas the succeeding Late Ordovician Kanling, Qiliang and Yin'gan formations have contrasting lithologies: the lower part (Kanling Formation) comprises largely interbedded layers of marls, mudstones and calcilutites; the middle part (Qiliang Formation) comprises mainly grey shales and mudstones, whilst the upper part (Yin'gan Formation) comprises mainly shales interbedded with siltstones and limestones (Fig. 2, Jia et al., 2004). In particular, the latest Ordovician integrative stratigraphy and timescale of China assigns the Saergan to Qiliang marine sediments in the Kalpintage region of Tarim as the deposits of the Darriwilian to Sandbian stages (~467–453 Ma, Zhang et al., 2019).

The Silurian-Devonian sequence in the Aksu and Kalpin areas succeeds the Upper Ordovician with disconformity and is assigned to the Kalpintage, Tataertage, Imogantau and Kezirtage formations (Fig. 2). In the early 1980s it was originally identified solely as the Devonian (Xinjiang Stratigraphic Compiling Group, 1981) although this assessment has been refined by more recent studies. The Kalpintage Formation is a shallow-marine to littoral facies composed of grey-green and purple red siltstones, mudstones, shales and fine-grained sandstones with a paleontological contact assigning deposition to the upper part to the Rhuddanian to early Aeronian stages of the Llandovery (Rong et al., 2003). The Tataertage Formation was deposited in a near-shore tidal flat environment and consists largely of purple-red and light grey siltstones, fine-grained sandstones, and mudstones containing brachiopods *Barroisella* sp. and *Lingula* sp. and gastropods *Naticopsis* sp. and *Eomphalus* sp., a paleontological assemblage assigning this formation to the Telychian stage of the Late Llandovery (Rong et al., 2003; Jia et al.,

2004). Deposition of the Imogantau Formation continued in a coastal tidal flat environment with purple-red sandy mudstones and siltstones sporadically interbedded with limestone lenses (Fig. 2) and a fauna including the conodonts *Ozarkodina excovata* and *Ozarkodina* sp., bivalves *Eurymyella* cf. *yunnanensis* and *Parllerodon* sp., and gastropods *Umbotropis albicans* and *Euomphalus* sp. assigning these strata to the Sheinwoodian to Homerian stages of the Wenlock. The succeeding Kezirtage Formation composed of sandstones, siltstones and claystones locally interrupted by conglomeratic interbeds records a fluviatile input from ancient meandering river courses. A paucity of fossils at this level precludes a precise age assignment although the lower part, comprising thick-bedded siltstones and fine-grained sandstones, was probably deposited during the Gorstian to Ludfordian stages (Jia et al., 2004). In summary, the Kalpintage and Tataertage formations are now both known to be of Llandovery age and the Imogantau and Kezirtage formations are of Wenlock, and Ludlow to Middle Devonian, ages respectively (Fig. 2). With an absence of significant intervening tectonism, these formations are in turn overlain disconformably by strata of Permian age (Fig. 2).

The Sishichang outcrops at Aksu (40.84°N, 79.83°E) dip to the southeast and are characterized by good exposure of an Upper Cambrian to Permian succession with a total thickness of ~2.5 km (Fig. 1b). For the present Ordovician investigation we have sampled twelve sites in the Saergan Formation (xk205–217) through a trench section at Subhash (40.53°N, 78.83°E) in Kalpin County where a Middle Cambrian to Permian sequence crops out to expose a ~2 km thick succession of Ordovician strata overlain disconformably by Lower Silurian rocks. The combination of a deep river cutting and the recent construction of a road has permitted the collection of pristine samples with clear bedding orientations in a region otherwise dominated by weathered outcrop.

For the Silurian study the (Llandovery) Tataertage and (Wenlock) Imogantau formations were selected for paleomagnetic sampling on the basis of quality of outcrop and clear definition of the formation boundaries at two regional localities. In the Subhash trench section the Llandovery Tataertage (ks83–87) and Wenlock Imogantau (ks77–82) formations 105 drill-core samples were collected from 11 sites, whilst in the Sishichang section a further 94 drill-cores at 10 sites from both the Tataertage (as35–40) and Imogantau (as41–44) formations in lithologies comprising purple to red fine-grained sandstones and mudstones (Table 2). The majority of the sampled Ordovician lithologies comprise sandy limestones and grey-green marls (Table 1 and Fig. 2) whilst the Silurian rock lithologies comprise mainly fine-grained sandstones (Table 2 and Fig. 2).

All the samples were extracted using a portable gasoline-powered drilling motor with drill-core samples orientated in situ mostly by sun compass, whilst some were orientated by both magnetic and sun compasses to evaluate differences between local magnetic declinations and the International Geomagnetic Reference Field (IGRF). As an example, the average difference between sun and magnetic compasses readings for the Ordovician and Silurian sediments from the Subhash trench section at Kalpin was 2.87° ± 0.36° ($n = 69$, 2σ) which conforms closely to the local magnetic field declination calculated from the IGRF model (~3°) for the sampling locality at 40.55°N, 78.91°E.

3. Rock magnetic results

Rock magnetic experiments comprising acquisition of isothermal remanent magnetization (IRM) accompanied by complementary back-field demagnetization, and progressive thermal demagnetization of composite IRMs (Lowrie, 1990) on the Ordovician pilot samples from the AKB area have identified either (i) a low coercivity ferromagnet such as magnetite with high coercivity magnetic minerals such as goethite and hematite barely detectable or absent, or (ii) a higher distributed acquisition spectrum with incomplete saturation achieved by 2700 mT (see Fig. 2 of Sun and Huang, 2009). In the former group, dominant low coercivity IRM behaviors with demagnetization signatures compatible with magnetite are found mainly in material from the mid-Ordovician Saergan Formation in the trench section at Subhash of Kalpin (sites ks89–92). The new Ordovician samples (sites xk205–217) were all collected from this section with sample lithologies comparable to sites ks89–92 (Table 1).

The anisotropy of magnetic susceptibility (AMS) measurements on specimens from the new Ordovician collection (xk205–217) identify predominantly oblate fabrics with corrected anisotropy degrees (P_j) ranging from 1.001 to 1.024 (Fig. 3a). Stereonet projections of the three principal axes of the AMS ellipsoids (Fig. 3b–c) show tilt-corrected k_{max} and k_{int} axial directions mainly distributed in the bedding plane and the k_{min} directions approximately perpendicular to bedding as anticipated from fabrics with a primary sedimentary origin. This AMS evidence implies that these rocks have escaped significant influence from the Cenozoic deformation and are thus more likely to have preserved a remanence acquired during quasi-primary sedimentary processes.

From the Silurian collection 22 pilot Silurian samples, one or two from each sampling site, were subjected to the rock magnetic measurements noted above. The majority of these samples fail to reach complete saturation of IRM by 2700 mT in a DC field (Fig. 4) indicating the presence of a dominant high-coercivity ferromagnet; some samples however also show a rapid increase of IRM in initial low fields of ~200–300 mT (e.g., Fig. 4a, c, d) identifying the additional presence of a low-coercivity magnetic mineral. The back-field demagnetization curves of SIRM show that most samples have remanent coercivities larger than 300–400 mT with a maximum as high as ~800 mT (Fig. 4). Clear acquisition spectral distinction between the two coercivity phases is present in samples such as ks83–9 (Fig. 4i) whilst a significant rapid decrease in SIRM during the initial back-field demagnetization is recognized for example, in as36–1 and ks77–10 (Fig. 4b, f). Thermal demagnetization of the composite IRMs shows a sample remanence composed predominantly of medium to hard components with unblocking temperatures of ~690 °C (Fig. 5), indicative of hematite; some samples show unblocking of a soft IRM component around ~550–580 °C (Fig. 5b, d, f, h–i) showing that magnetite is present as a subordinate ferromagnet; a minority of samples show rapid drop of remanence of hard and medium IRM components during initial stages of thermal demagnetization (Fig. 5b) indicating that goethite is also present in some of the Silurian sandstone samples.

4. Paleomagnetic results

In the context of the range of ferromagnets resolved from the rock magnetic behaviors of pilot samples (Sun and Huang, 2009 and Figs. 4–5) identifying magnetite and high- and low-unblocking temperature high coercivity ferromagnetic minerals in different portions of this rock collection, a two-fold demagnetization strategy of ~9 specimens per site was adopted. All Ordovician specimens were subjected to hybrid demagnetization comprising first progressive thermal demagnetization up to 250 °C or 300 °C, and then alternating field (AF) demagnetization to a peak field of ~100 mT. This procedure was adopted to remove the contribution of low temperature but high coercivity remanence likely acquired during superficial weathering but appeared to be unnecessary for the Silurian collection which was accordingly treated in the conventional way by progressive thermal demagnetization. For the thermal demagnetization, the specimens were degaussed using a TD-48 thermal demagnetization oven with temperature intervals of 50 to 100 °C adopted in lower temperature ranges and 10 to 30 °C in higher ranges. The AF demagnetization was performed using a 2G600 AF degausser coupled to the 2G755R cryogenic magnetometer with a general peak field interval of 5–10 mT. Remanence measurements were conducted on a 2G755R cryogenic superconducting magnetometer in the Paleomagnetism and Geochronology Laboratory of the Institute of Geology and Geophysics, Chinese Academy of Sciences, where all degaussers and magnetometers are

Table 2
Summary of Silurian paleomagnetic results for characteristic remanence component from the Aksu-Kalpin areas of northwestern Tarim.

Site ID	Lithology	Location		Bedding	n/n_0	R/N	In-situ		Tilt-corrected		k	α_{95}(°)	VGP	
		φ_s(°E)	λ_s(°N)	Strike/dip			D_g(°)	I_g(°)	D_s(°)	I_s(°)			φ_p(°E)	λ_p(°N)
Llandovery Tataertage formation														
as35[b]	Quartz sandstone	79.8331	40.8327	70/59	5/8	0/5	80.0	31.3	101.8	8.1	20.7	17.2	159.0	−6.2
as36[a,b]	Quartz sandstone	79.8326	40.8325	69/59	4/9	0/4	159.9	53.2	159.5	−5.8	22.7	19.7	111.2	−47.8
as37[a,b]	Quartz sandstone	79.8327	40.8323	68/56	7/8	1/6	74.1	63.4	127.9	27.4	4.9	30.2	132.7	−16.6
as38[b]	Quartz sandstone	79.8327	40.8323	66/56	0/9	No meaningful site mean could be calculated								
as39[b]	Quartz sandstone	79.8329	40.8321	66/54	7(2)/9	0/7	59.6	72.9	134.8	36.0	45.9	9.0	123.8	−16.1
as40[b]	Sandstone	79.8333	40.8321	69/51	6(3)/8	5/0	227.0	−30.1	262.0	−34.7	38.0	11.0	340.0	−18.3
ks83	Sandstone	78.8635	40.5368	216/45	9/9	0/9	63.1	−12.9	64.5	9.1	61.8	6.6	182.5	22.2
ks84	Sandstone	78.8635	40.5368	216/45	9/9	0/9	70.3	−19.0	73.8	8.4	244.2	3.3	176.3	15.0
ks85	Sandstone	78.8637	40.5368	218/41	8/9	0/8	78.8	−6.2	74.8	20.2	75.8	6.4	170.7	18.3
ks86	Sandstone	78.8638	40.5369	218/41	7/9	0/7	74.0	0.4	66.6	23.0	82.1	6.7	174.9	25.5
ks87	Sandstone	78.8640	40.5369	221/43	7/9	0/7	70.9	−9.8	70.1	12.1	64.0	7.6	177.3	19.1
Wenlock Imogantau formation														
as41[b]	Sandstone	79.8344	40.8318	68/58	9/9	0/9	68.1	52.5	115.9	24.8	28.6	9.8	142.7	−10.1
as42[b]	Sandstone, mudstone	79.8348	40.8312	68/61	9/9	0/9	57.1	37.5	98.5	25.2	33.7	9.0	154.3	2.3
as43[b]	Sandstone, mudstone	79.8349	40.8311	68/61	6(3)/8	0/6	49.9	54.3	116.3	33.5	56.4	9.0	138.8	−6.5
as44[b]	Sandstone	79.8349	40.8311	67/55	8/8	0/8	59.3	44.8	103.6	28.8	102.4	5.5	149.4	0.1
ks77	Sandstone	78.8335	40.5279	240/35	4/9	4/0	273.0	−16.0	259.0	−31.8	159.3	7.3	342.3	−19.3
ks78[a]	Sandstone	78.8340	40.5279	233/31	5/9	5/0	288.0	−6.5	281.0	−31.1	5.9	34.5	329.0	−2.8
ks79	Sandstone	78.8345	40.5273	227/36	4/9	4/0	275.0	−8.9	265.0	−34.0	73.3	10.8	337.5	−15.7
ks80	Sandstone	78.8354	40.5265	231/36	6/9	0/6	81.0	10.7	69.7	26.0	16.9	16.8	171.3	24.2
ks81	Sandstone	78.8620	40.5371	229/36	9/10	0/9	68.4	−22.9	77.7	−7.8	28.0	9.9	179.9	6.7
ks82	Sandstone	78.8622	40.5369	229/45	9/10	0/9	75.6	0.7	68.0	19.0	44.5	7.8	175.8	23.0
GMA	All sites				17/21	3/14	71.6	17.0			7.1	14.4	162.0	10.5
									86.1	22.9	13.4	10.1	K = 15.1	A_{95} = 9.5
GMB	Sites without those marked by “a” and “b”				10/21	2/8	77.0	−3.5			24.3	10.0	172.9	19.1
									72.7	17.7	34.6	8.3	K = 78.1	A_{95} = 5.5
GMC	Sites marked by “b” only				7/21	1/6	60.4	46.7			22.2	14.6	147.0	−2.7
									107.3	28.2	22.5	13.0	K = 23.5	A_{95} = 12.7

Fold test:
(1) Watson and Enkin's test (1993): GMA: N = 17, the optimum concentration is achieved at 86.7 ± 5.3% unfolding.
(2) McFadden's test (1990): GMA: N = 17, in-situ ξ_2 = 13.466, tilt-corrected ξ_2 = 3.091, statistical threshold ξ = 4.801 at 95% confidence level, indicative of a positive fold test.
GMB: N = 10, in-situ ξ_2 = 4.357, tilt-corrected ξ_2 = 1.868, statistical threshold ξ = 3.685 at 95% confidence level, indicative of a positive fold test.
Fisherian group mean directions calculated following correction of sampling site locations to the Kalpin Subhash section at 78.9°N, 40.5°N. The remaining symbols are as for Table 1.

[a] Sites discarded duo to large uncertainties (α_{95} > 18°).
[b] Site-mean directions of characteristic remanence interpreted to be contaminated by regional Permian igneous activity.

installed in a field-free space. Demagnetization results were analyzed using principal component analysis (Kirschvink, 1980) and interval-mean results evaluated from Fisher statistics (Fisher, 1953), or by a combination method of great-circle fits (Halls, 1978) and directional observations (McFadden and McElhinny, 1988).

4.1. The Ordovician

The hybrid thermal and AF demagnetization of Ordovician limestone and marl specimens illustrates demagnetization trajectories of two remanent magnetization components in the majority of specimens (Fig. 6a–e) comprising a low-temperature component (LTC) resolved below ~250 °C and a high field characteristic remanent magnetization (ChRM) identified between alternating peak fields of 20–30 mT and ~60–90 mT. The LTC is of single normal polarity and grouped close to the Present Geomagnetic Field direction (D/I = 357°/59°) before tilt correction with obvious deterioration in data grouping after tilt correction. These two observations define this component as a recent viscous overprint or Late Cenozoic secondary remanence. Some limestone specimens exhibit a high field demagnetization trajectory closely following a great-circle (Fig. 6c–e) permitting component isolation by remagnetization great-circle fitting (Halls, 1978). Similar to the comparable early Paleozoic lithologies investigated in earlier studies (e.g., Huang et al., 1999; Sun and Huang, 2009), some limestone specimens exhibit only an erratic high field demagnetization trajectory (e.g., specimens from sampling sites xk211, 213, 215 and 216, Table 1) where no stable ChRMs could be isolated. As a result, 73 out of 114 demagnetized specimens yielded either a stable ChRM direction or a high field best fitting great-circle demagnetization trajectory (Fig. 6a–e). The ChRM directions are dominated by normal polarity with quasi-antipodal directions being observed mostly at sampling sites xk205–206 (Table 1). However, ChRM directions from site xk207 are quite different from other sampling sites both before and after tilt correction (Table 1) and are excluded from this ChRM group. The remaining 65 ChRM observations yield a formation mean of D = 44.3°, I = 61.8° (k = 21.3, α_{95} = 3.9°) before and D = 102.1°, I = 50.7° (k = 24.2, α_{95} = 3.7°) after tilt adjustment. A reversal test (McFadden and McElhinny, 1990) yields a negative result (the angular distance between the normal and reversed ChRM means is γ_0 = 24.4° ± 5.0° with a critical angle of γ_c = 6.6° at 95% confidence level), a result which reflects the small number but concentrated distribution of the reversed ChRM observations. A paleomagnetic fold test is unfortunately not practical on these newly-isolated ChRM observations due to the location of the Subhash trench section (Fig. 1b) on the common limb of a monoclinal structure.

4.2. The Silurian

All Late Llandovery and Wenlock specimens of fine-grained sandstone and mudstone lithologies were subjected to thermal demagnetization up to ~690 °C in the context of the high coercivity hematite observed as the dominant magnetic mineral from the rock magnetic

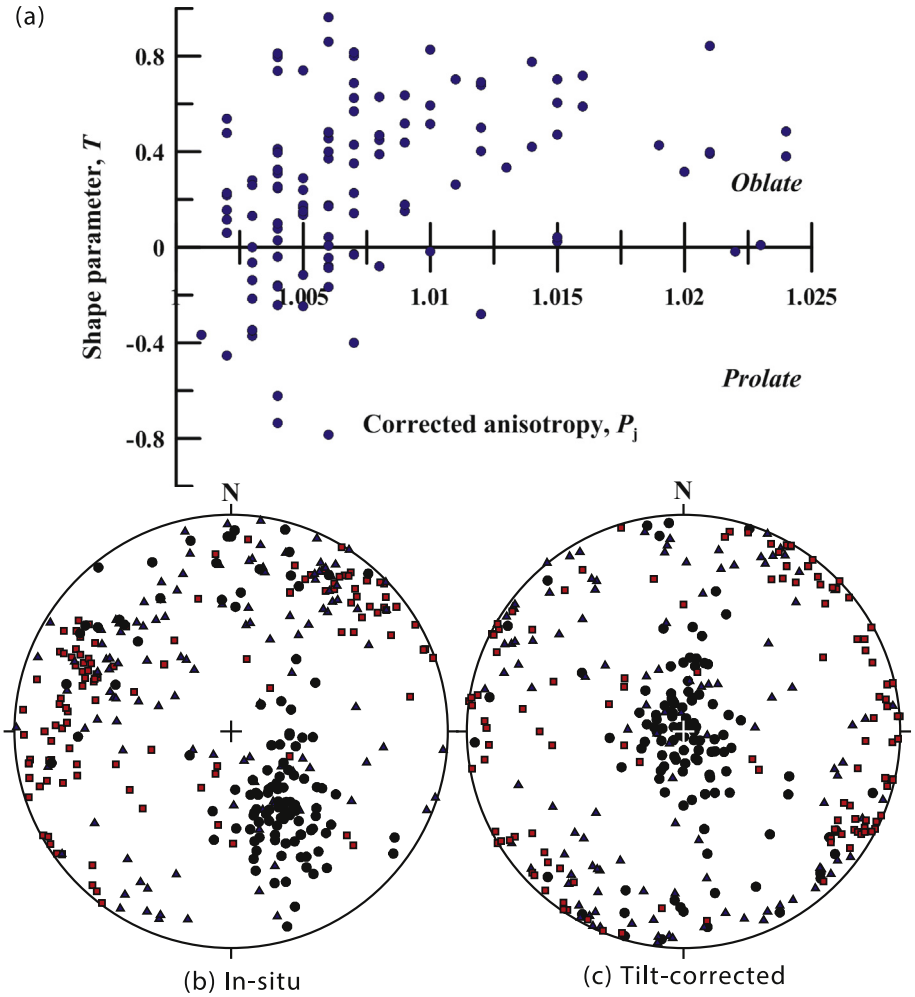


Fig. 3. Results from anisotropy of magnetic susceptibility (AMS) measurements on 117 limestone and marl specimens collected from the Middle-Late Ordovician Saergan Formation of Kalpin Subhash trench section (sites xk205–217). (a) Corrected anisotropy degrees (P_j) of magnetic susceptibility versus shape parameter (T); (b–c) lower hemisphere stereonet projections of three principal axes of the AMS ellipsoids where squares, triangles, and circles indicate the maximum, intermediate, and minimum principal axes respectively.

investigations (Figs. 4–5). Demagnetization of 208 Silurian specimens identified two components in most specimens following removal of a viscous component during initial stages of demagnetization at temperatures of ~100–200 °C (Fig. 7). A LTC was isolated in some 165 of the 208 specimens by treatment to temperatures of ~300–400 °C. As shown in Supplementary Fig. S1, all these component directions are distributed around the Present Geomagnetic Field (PGF) in geographic coordinates with a Fisherian (Fisher, 1953) mean direction of D/I = 1.9°/56.3° (α_{95} = 1.9°) and show significant deterioration of directional grouping following tilt adjustment (k_s/k_g = 0.08). Again, these two observations identify this LTC as a late acquisition in the Cenozoic geomagnetic field.

Higher levels of demagnetization define trajectories showing a straightforward decay to the origins of the vector projections and identify a high temperature ChRM unblocked by temperatures of 650–680 °C in the majority of specimens (Fig. 7a–d, h, j) although removed by ~590 °C in a few cases (Fig. 7g), and consistent with the rock magnetic properties observed in these Silurian samples (Figs. 4–5). Some specimens, notably from the Tataertage Formation of the Aksu Sishichang section (sites as35–40), showed no systematic decay of remanence to the origin (Fig. 7e–f, i) and failed to yield a high temperature component from principal component analysis. However, amongst these samples a few specimens from sites as39 and 40 exhibit great circle migration on an equal-area projection permitting determination of ChRM directions from a combination of great circle fits and directional observations (Table 2).

As noted in Table 2, two Llandovery sampling sites (as36, 37) from Sishichang and one Wenlock sampling site (ks78) from Subhash show problematic site-mean directions with uncertainties (α_{95}) larger than 18°. Excluding these 3 questionable observations, site-mean observations of the ChRM could be confidently isolated from eight Llandovery and nine Wenlock sites in both Sishichang and Subhash sections. Tilt-corrected ChRM directions have predominately northeast-to-east declinations and shallow downward inclinations, with an antiparallel equivalent at one Llandovery (as40) and two Wenlock (ks77, 79) sites. Collectively the 17 site ChRM directions yield a group mean of D = 71.6°, I = 17.0° (α_{95} = 14.4°) before, and D = 86.1°, I = 22.9° (α_{95} = 10.1°) after, tilt adjustment (Fig. 8a–b) with directional clustering significantly improved (k_s/k_g = 1.89) and an optimal grouping achieved with a 86.7 ± 5.3% level of unfolding. Further application of the McFadden (1990) fold test also yields a positive result at the 95% confidence level; the statistic ξ_2 is 13.466 before and 3.091 after tilt correction with a critical threshold of ξ = 4.801 at this confidence level. Unfortunately the field constraints noted above only permit a pre-Cenozoic age to be assigned to this remanence from the evidence of the fold test alone.

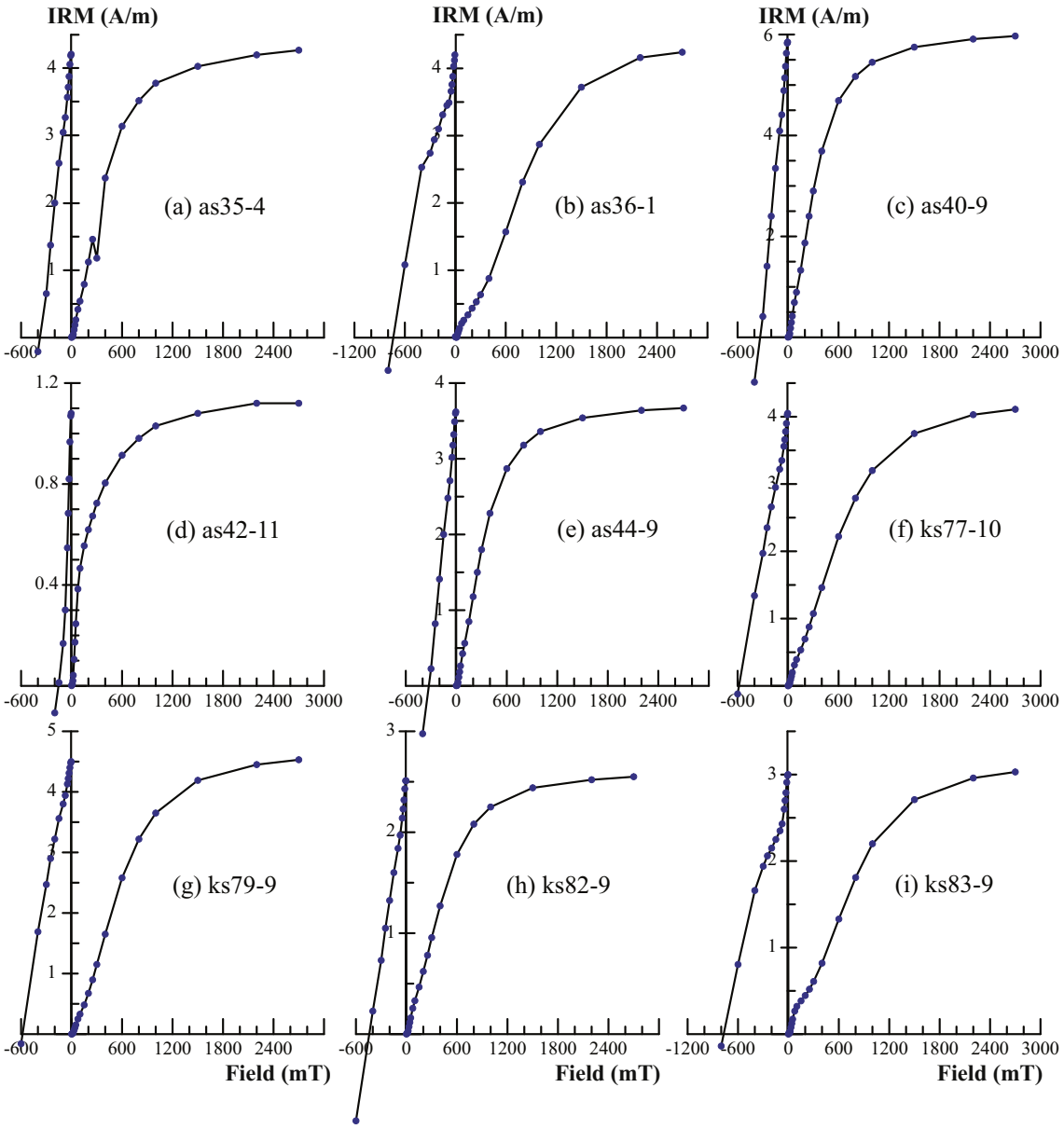


Fig. 4. Behavior of isothermal remanent magnetization (IRM) acquisition and back-field demagnetization of saturation IRM for pilot Silurian samples. The IRMs were acquired in a direct current (DC) magnetic field using a 2G660 pulse magnetizer and measured by JR-5 spinner magnetometer in the Paleomagnetism and Geochronology Laboratory of the Institute of Geology and Geophysics, Chinese Academy of Sciences.

5. Discussions

5.1. Origin of the characteristic remanent magnetizations

5.1.1. The Ordovician

Of the full collection of Ordovician cores collected from 57 sampling sites in the AKB area (this study and Sun and Huang, 2009) most specimens were found to carry a stable ChRM: the exceptions were some magnetically-weak limestone and dolomitic limestone specimens collected from the Early to Middle Ordovician Shangqiulitage Formation (sites aq56–61 and ks94–99) of the Aksu and Kalpin areas. The stable ChRMs group into two categories separated by application of tilt adjustment (Fig. 9 and Supplementary Fig. S2). The first group, designated component A hereafter, comprises 29 site observations from our previously-reported study and the site-mean direction of xk207 (Supplementary Table S1); *in-situ* these magnetizations are directed mainly NW-to-NE with intermediate to steep downward inclination and near-antiparallel equivalents (Supplementary Fig. S2a). The A component is mainly isolated from limestone (sometimes argillaceous) and marl specimens with hematite and subordinate magnetite as magnetic carriers, and is apparently independent of rock formation age. Fold test analysis (McFadden, 1990) of 30 site-mean directions shows a significantly-negative result with an optimal grouping achieved by 3.9 ± 1.8% unfolding (Supplementary Table S1) and strongly implies that this component (A) postdates the regional folding. In the context of the negative fold test and the similarity of the corresponding paleomagnetic pole prior to tilt adjustment (84.8°N and 113.3°E, A_{95} = 5.7°) to the Cenozoic reference field from the Tarim Block (e.g., Fang et al., 1998), we interpret the A component as a Cenozoic overprint linked to the extensive deformation of this age in central Asia.

The second group of ChRM directions is found where magnetite is evidently the dominant carrier of remanence (component B) but proves to be resolvable only in the Middle-Late Ordovician Saergan Formation from the Subhash trench section and 7 specimens in the Late Ordovician Qiliang Formation from the Sishichang section (Table 1 and Figs. 6, 9). This ChRM was usually isolated at relatively higher alternating fields by

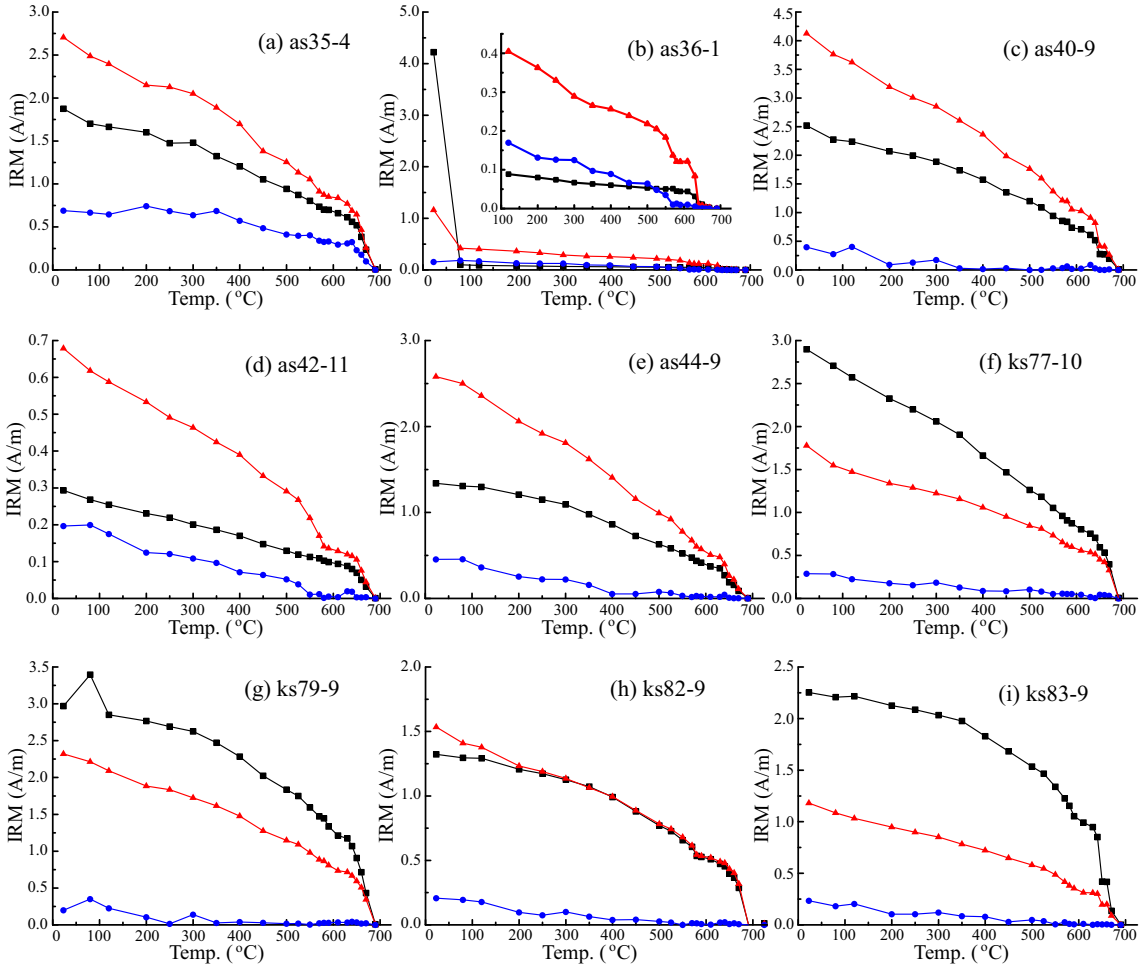


Fig. 5. Progressive thermal demagnetization curves for composite IRMs of pilot Silurian samples. The composite IRMs were applied in DC magnetic fields of 50, 500 and 2700 mT to the sample X-axis, Y-axis and *Z*-axis respectively. Progressive thermal demagnetization of the composite IRMs was carried out using an ASC-TD48 oven and remanence was measured with a 2G Enterprises cryogenic magnetometer. The circles, triangles and squares identify soft, medium and hard IRM components, respectively.

the hybrid thermal and AF demagnetization sequence following complete removal of the LTC (Fig. 6). Demagnetization trajectories in some specimens showed great circle migration on an equal-area projection following LTC removal (Fig. 6c–e, g, i) permitting site-mean ChRM determination either from intersection of great circles or from a combination of great circles with directional observations. The directional grouping is markedly improved following tilt adjustment at both sampling site and specimen levels (Fig. 9). The positive fold test results (k_s/k_g = 23.7/8.1 and Table 1) suggest that the B component was acquired before the regional folding. As noted in Section 2, the Phanerozoic successions in the Aksu-Kalpin areas of northwestern Tarim are characterized by the absence of Late Devonian to Carboniferous and Mesozoic rocks (Fig. 2 and Jia et al., 2004). The Permian strata have a disconformable relationship with the underlying Middle Devonian rocks (Fig. 2) indicating that only mild deformation occurred during the Upper Devonian-Carboniferous interval. In contrast, the Cenozoic rocks are in contact with the Paleozoic and older rocks with marked unconformity and define late Mesozoic-early Cenozoic times as the interval embracing major deformation in this region (e.g., Jia et al., 2004). Hence the positive fold test results for component B are only able to confidently define an origin for this ChRM predating this last deformation. However, a marked difference between the corresponding tilt-corrected paleomagnetic pole (33.7°S, 185.0°E, dp/dm = 2.7°/4.0°) and well-defined Permian poles from volcanics and dykes (Li et al., 1988; Sharps et al., 1989) as well as the wider post-Silurian apparent polar wander path (APWP) from the Tarim Block (Fig. 10, Table 3 and Huang et al., 2018), indicates that the B component is very likely to be a primary remanence acquired at, or close to, the formation age of these rocks. Thus we assign a median age of ~460 Ma to this Middle-Late Ordovician paleopole, which is resolved from the Darriwilian Saergan and the Sandbian Qiliang formations (Zhang et al., 2019).

On the other hand, the rock magnetic comparison between specimens bearing the A and B components shows that the secondary A component is linked to high coercivity remanence because the predominant ferromagnet in these argillaceous limestone and marl samples is hematite and possibly goethite, with the presence of only a subordinate magnetite signature. Whilst AMS results indicate that ferromagnet growth was constrained to depositional fabrics (Fig. 3), it has evidently enhanced the NRM intensity so this late secondary chemical remanent magnetization (CRM) has partially or completely overprinted any detrital remanent magnetization (DRM) carried by primary or early-diagenetic magnetite. It is recognized that diagenetic processes in carbonate rocks can sometimes result in formation of authigenic pyrrhotite, maghemite or pyrite, (with the former two being ferromagnets of moderate to low unblocking temperatures); this possibility is embraced by the treatment procedure adopted and may be consistent with some demagnetization results at lower temperatures (see also Huang et al., 1999). Nevertheless temperature spectra and directional properties of the ChRMs are distinct and compatible with interpretation of the B remanence as a probable quasi-primary acquisition; in contrast

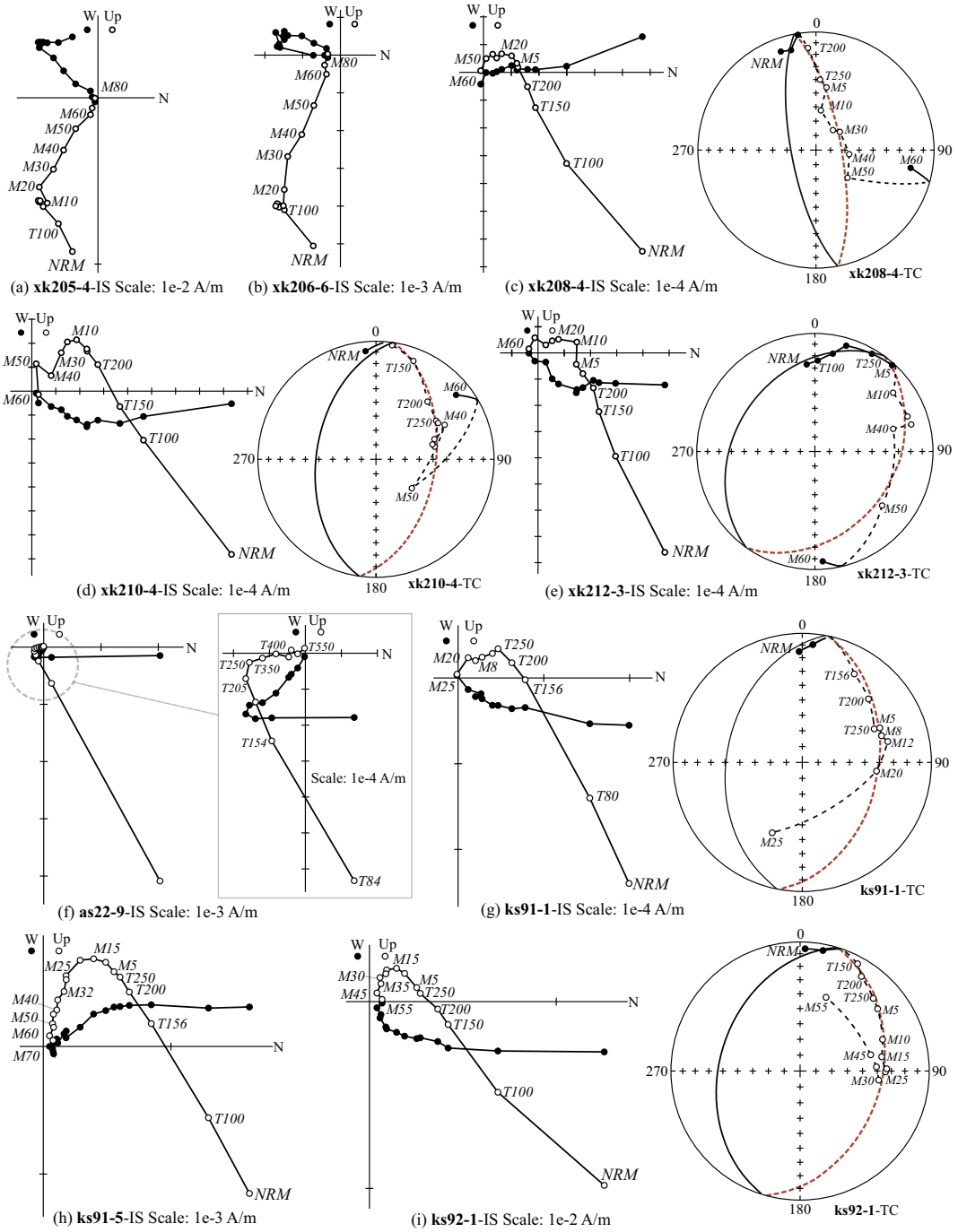


Fig. 6. Orthogonal vector plots (Zijderveld, 1967) of representative Ordovician specimens and equal-area projections of progressive demagnetization results for representative pilot specimens. Solid and open circles in orthogonal plots (all drawn *in-situ*) indicate vector endpoints projected onto the horizontal and vertical planes, respectively. Solid and open circles in equal-area projections (all plotted after tilt adjustment) represent directions plotted onto lower and upper hemispheres, respectively. T and M refer to thermal (°C) and alternating field (mT) demagnetization, respectively. Figures f–i are redrawn from Sun and Huang (2009).

the dual polarity but post-folding A remanence is likely to be the signature of much later hydrothermal processes precipitating secondary hematite and operating during the extensive Cenozoic tectonism in central Asia.

5.1.2. The Silurian

The Late Llandovery and Wenlock rocks from the Aksu and Kalpin areas carry high temperature ChRM components in the majority of the sampled fine-grained sandstone and mudstone facies. Both collections yield positive fold tests (McFadden, 1990; Watson and Enkin, 1993) and the 17 site-mean directions identified from these high temperature ChRMs have NE-to-E declinations and shallow downward inclinations with a few antiparallel components. However, a distinct directional deviation toward a southeast declination is recognized in the majority of site-mean directions from the Sishichang section (Fig. 8b). The collective evidence relevant to interpreting the origin of this latter declination swing comes from a range of studies; these comprise 20 sites in the (Wenlock) Imogantau and (Gorstian to Givetian) Kezirtage formations in the Aksu Sishichang section (Bai et al., 1987), 14 sites from the upper part of the (Llandovery to Givetian) sequence in the Kalpin Yin'gan section (Li et al., 1990), 6 sites from a fraction of the (Llandovery to Givetian) sequence in the Kalpin Acha section, 42

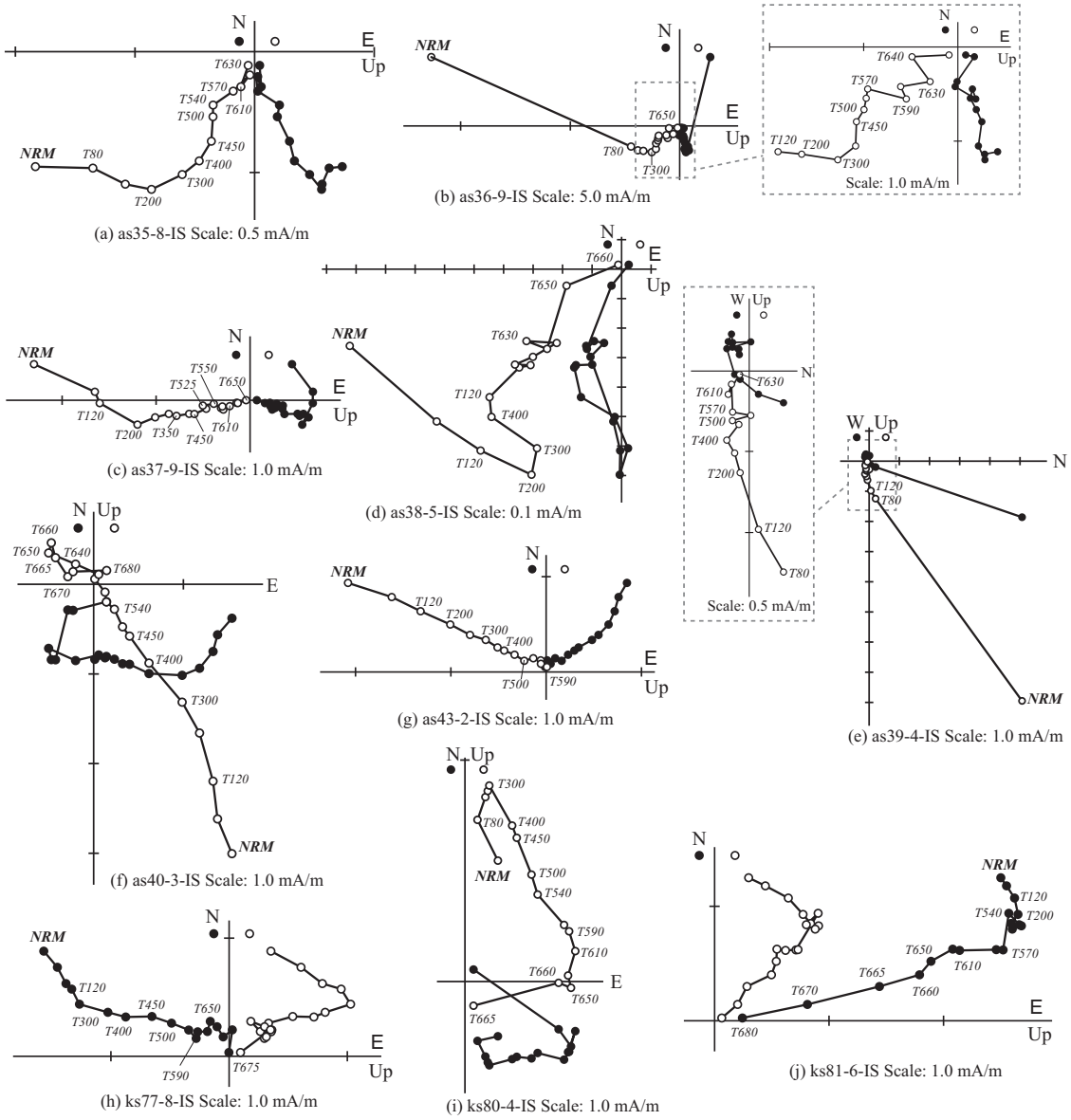


Fig. 7. Orthogonal vector plots (Zijderveld, 1967) of representative Silurian specimens. The remaining symbols are as for Fig. 6.

magnetostratigraphic sites from a 1998 m (Llandovery to Givetian) sequence in the Kalpin Subhash section (Li et al., 1990) and 4 sites from the (Wenlock) Imogantau Formation in the Aksu Sishichang and Kalpin Yin'gan sections (Fang et al., 1996, see also Table 3). Analysis on a section by section basis of pre-folding (i.e. pre-Late Mesozoic to Cenozoic) paleomagnetic poles from each section finds that poles from the Sishichang and Yin'gan sections lie on a plane defined by the (reversed polarity) Permian pole (Sharps et al., 1989) and the Subhash Silurian to Devonian pole (Li et al., 1990). This indicates that the Silurian to Devonian ChRM directions from these two sections are biased toward the paleofield directions resolved from basaltic lavas and dykes of the extensive regional Permian magmatism (see Li et al., 1990 and references therein). In the context of this observation we note that 7 virtual geomagnetic poles (VGPs) derived from Llandovery to Wenlock rocks of the Aksu Sishichang section are biased toward this reversed polarity Permian pole (Fig. 8c) implying that the southeastward deviating site-mean directions found at these sites record contamination by the regional Permian igneous activity during the Kiaman Superchron.

Accordingly we interpret 10 of the Silurian site-mean directions where this signature is absent in the context of a likely primary remanence. These yield a group mean of $D = 77.0^{\circ}$, $I = -3.5^{\circ}$ ($\alpha_{95} = 10.0^{\circ}$) before and $D = 72.7^{\circ}$, $I = 17.7^{\circ}$ ($\alpha_{95} = 8.3^{\circ}$) after tilt correction (Table 2). Data grouping is monotonically improved with increasing unfolding; however the ratio of precision parameter before and after the tilt adjustment ($k_s/k_g = 1.42$) is not significant at the 95% confidence level due to only small variations in bedding attitudes in the Subhash section. Nevertheless application of the McFadden (1990) fold test yields a positive fold test result with statistics of $\xi_2 = 4.357$ before and 1.868 after tilt correction with a critical threshold of $\xi = 3.685$ at the 95% confidence level. Specifically this pole has evidently escaped the influence of the Permian igneous activity in Tarim and is significantly removed from any known Permian and younger paleomagnetic poles from the Tarim Block (Table 3 and Huang et al., 2018). We therefore consider that the high temperature ChRM isolated from the Llandovery and Wenlock rocks in the Subhash section was acquired prior to the Kiaman reversed superchron and, because no major thermal/tectonic events are evident between Silurian and Permian times, is likely to be a remanence resulting from primary lithification and diagenetic processes. We therefore propose that the corresponding paleomagnetic pole at 19.1°N, 172.9°E ($K = 78.1$, $A_{95} = 5.5^{\circ}$) can be used to constrain the location of the Tarim Block during the deposition interval of the Late Llandovery (Telychian) Tataertage and Wenlock

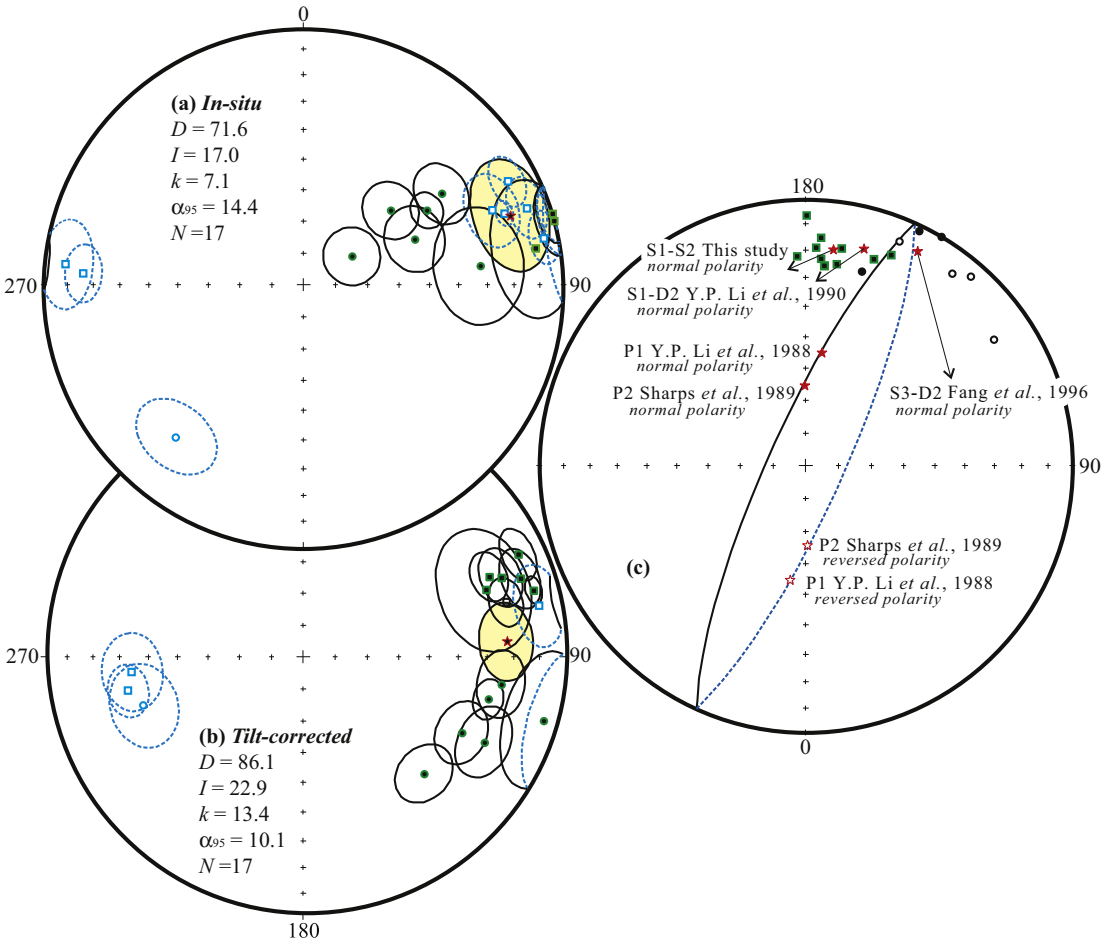


Fig. 8. (a–b) Equal-area projections of site-mean directions of high temperature characteristic remanence isolated from Silurian rocks from the Aksu Sishichang (circles) and Kalpin Subhash trench (squares) sections of northwestern Tarim before (a) and (b) after tilt correction; solid and open symbols represent directions plotted onto lower and upper hemispheres, respectively. (c) Equal-area stereographic projections of Silurian, Devonian, and Permian paleomagnetic poles from Tarim with a great circle through the Permian poles in both normal and reversed polarities. Solid (open) symbols of corresponding virtual geomagnetic poles for tilt-corrected site-mean directions from the Aksu Sishichang (circles) and Kalpin Subhash (squares) sections plotted on the lower (upper) hemisphere.

Imogantau formations (~439–427 Ma, Table 3).

5.2. Paleogeography of the Tarim Block during the Early Paleozoic

In a recent analysis of paleomagnetic evidence for the paleogeographic evolution of East Asia (Huang et al., 2018) we quoted results from the present study in preliminary form to extend the record from Tarim back into the Paleozoic Era beyond the anchor pole provided by the well-established Permian record. From the report of the primary evidence reported in this paper we now develop the case for establishing the kinematic record of Tarim prior to the Permian by justifying the quasi-primary origin of the magnetizations and establishing their polarity. The evidence for the quasi-primary origin described above is fivefold; in the order of greater to lesser significance these are: (i) the removal of these pole positions from younger paleomagnetic results so far reported from Tarim (e.g., Huang et al., 2018), (ii) their compatibility with the previous limited Ordovician-Silurian record from this block (e.g., Li et al., 1990; Fang et al., 1996; Zhao et al., 1997), (iii) the fold test establishing a pre-Cenozoic age, (iv) the AMS record suggesting a primary sedimentary fabric reinforced by the absence of major regional thermal-tectonic activity in the study areas prior to Cenozoic times, and (v) the definable, but partial, record of overprinting by the regional Permian igneous activity in these older rocks. The Ordovician B component is derived from a group magnetization dominated by E-to-SE declinations and upward inclinations following tilt adjustment (Table 1 and Fig. 9). The geomagnetic field during the Late Darriwilian epoch of the Middle Ordovician and ensuing Late Ordovician incorporated the Sandbian–Early Katian interval of mixed polarity whilst the Late Katian–Hirnantian interval was dominated by normal polarity (Pavlov and Gallet, 2005; Gradstein et al., 2012). By applying the shortest polar wander principle to the APWP construction, we therefore interpret the dominant E-to-SE declinations and upward inclinations of this B component as the record of a normal polarity field. This new Mid-Late Ordovician paleomagnetic pole, with a median age of ~460 Ma, would then indicate that Tarim was sited in moderate Southern Hemisphere paleolatitudes, specifically 32.4°S for the Kalpin reference site (40.5°N, 79°E). Furthermore the result merits comparison with preliminary paleomagnetic results from Late Cambrian and Middle Ordovician limestones in the Kuruktage area of northeastern Tarim reported by Zhao et al. (1997) which show comparable paleolatitudes of 27.8°S and 36.5°S during Late Cambrian and Middle Ordovician times respectively. The collective evidence therefore implies that the Tarim Block was subject to only small latitudinal motions in moderate southerly paleolatitudes during the Cambro-Ordovician interval. In marked contrast, comparison between the Mid-Late Ordovician paleomagnetic pole and the probable Late Llandovery-Wenlock Silurian paleopole (~433 Ma) suggests large northward movement of the Tarim Block defined by the present study as 41.9° ± 4.7° accompanied by a clockwise vertical-axis rotation of 35.6° ± 5.0°. This rapid northward displacement would have translated the Tarim Block across the paleo-

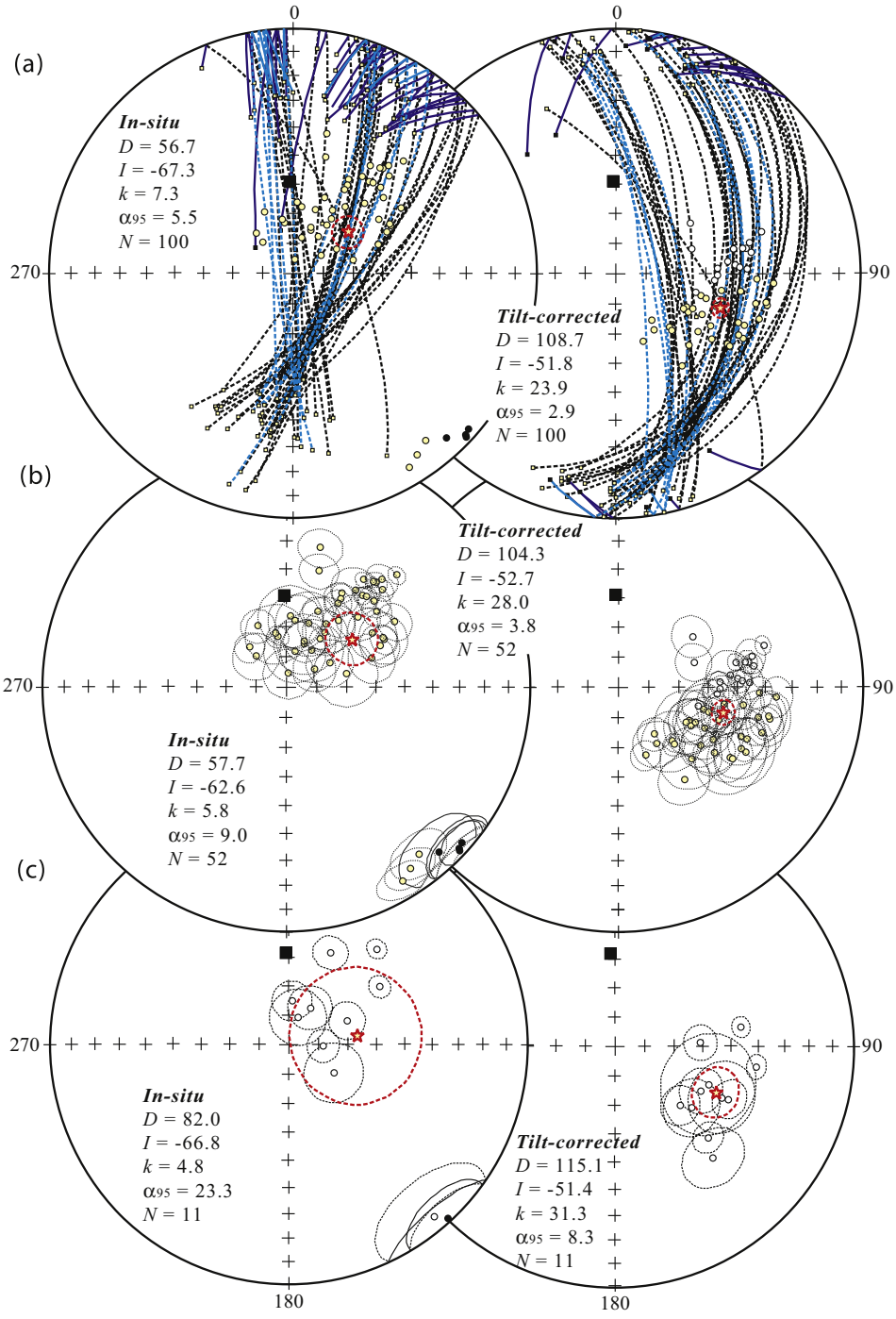


Fig. 9. Equal-area projections of directions of the characteristic component B isolated from Ordovician rocks before (left) and after (right) tilt correction. (a) Specimen level group mean directions calculated from a combination of 48 great-circle fits and 52 directional observations; (b) specimen level group mean directions calculated from 52 directional observations only; (c) site level group mean directions for all the 100 specimens. Stars show group mean directions with solid and open symbols recording directions plotted onto the lower and upper hemispheres, respectively.

equator into low northerly paleolatitudes of ~9°N between ~460 and 433 Ma.

Table 3 summarizes the present paleomagnetic record from the Tarim Block for the late Neoproterozoic to end Paleozoic interval. The paleopoles listed here are interpreted to be of normal polarity with the possible exception of the three Neoproterozoic poles because large, and yet poorly defined, APW motions are identified between mid-Ediacaran and mid-Cambrian times (Torsvik et al., 2012; Piper, 2013). The first key feature of the subsequent path is the large northward motion of the pole noted above and constrained broadly to the interval between 467 and 427 Ma with the earlier anchor of this motion having ancillary support from the reconnaissance study of Fang et al. (1996). Following the ~35° latitudinal motion recognized here and focused within the whole or part of the Late Ordovician to Late Silurian interval, there appears to have been little subsequent motion of Tarim until a large motion carried the pole into high latitudes between 383 and 359 and 359–299 Ma in a rapid motion likely concentrated within Early Carboniferous times (Huang et al., 2018). The pole then remained sited in intermediate to high present latitudes evidently recording minimal drift through the remainder of Paleozoic times (Table 3).

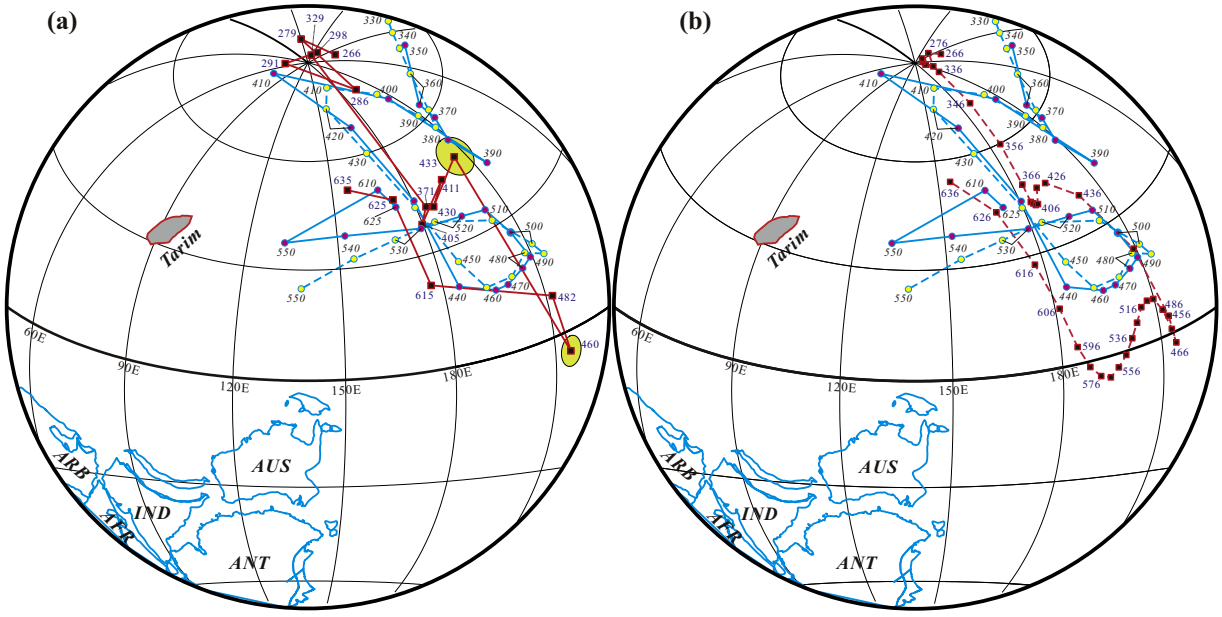


Fig. 10. Schmidt projection of late Neoproterozoic to Paleozoic apparent polar wander (APW) paths in Australian coordinates showing a comparison between Gondwana and Tarim. The late Neoproterozoic to Paleozoic APW paths for Gondwana include late Neoproterozoic (625 and 610 Ma) paleomagnetic poles from Australia (Schmidt et al., 1991; Schmidt and Williams, 1995; Sohl et al., 1999) and the spherical spline Paleozoic APW paths of Torsvik et al. (2012) with (dashed line) and without (solid line) potential inclination error correction for those from sedimentary rocks. Inputs for the Tarim Block include late Neoproterozoic-Guadalupian paleomagnetic poles listed in Table 3 where the 329, 286, 279 and 266 Ma paleopoles are Fisherian mean poles calculated from individual poles during the intervals noted (a). The red dashed line is a spherical spline APW path for the Tarim Block with smoothing factor = 300 (b). The Tarim Block and APW paths (a, b) are rotated into Australia coordinates about an Euler pole at 45.9°S, 277.6°E with an angle of −43.4° (P. Zhao et al., 2014). The fit of Gondwana is plotted according to Euler rotation parameters in Torsvik et al. (2012). Blocks shown in this figure and/or Fig. 11 are abbreviated: AFR, Africa; ANT, Antarctica; ARB, Arabia; AUS, Australia; BAL, Baltica; G, Greenland; IND, India; KAZ, Kazakhstania; M, Madagascar; SAM, South America; SIB, Siberia. The two new early Paleozoic paleopoles for Tarim are highlighted by their 95% confidence limits. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

The key challenge to defining motions of Tarim in common with the other blocks now comprising central-east Asia is to isolate the intervals when they were moving in contiguity from times when they moving as separated blocks. Similar paleolatitude trends have been recognized in the Tarim, South China and Australia blocks during the Neoproterozoic to early Paleozoic interval (e.g., Yang et al., 2004; Huang et al., 2005, 2018; Zhao et al., 2014; Han et al., 2015). However the new early Paleozoic paleomagnetic data reported here do not support a long-term late Neoproterozoic to mid-Silurian Australia-Tarim connection (Chen et al., 2004; Zhan et al., 2007; Zhao et al., 2014). Thus when the early Paleozoic APWP of the Tarim Block is rotated according to the Tarim to Australia coordinates of Zhao et al. (2014) about an Euler pole at 45.9°S, 277.6°E with an angle of −43.4°, the Tarim path is observed to be far removed from the APWPs for Gondwana in Australian coordinates (Torsvik et al., 2012 and Fig. 10), an observation that appears to negate any putative long-term late Neoproterozoic to mid-Silurian connection between Australia and Tarim (Zhao et al., 2014).

The latitudinal motion and clockwise rotation of Tarim defined here during ~460 to 433 Ma interval merits comparison with the polar migration shown by the fitted spline APW path for Laurentia between ~460 and 430 Ma (Torsvik et al., 2012). This later reference frame defines an APW motion which is larger, and latitudinal motion which is appreciably less, than Tarim with the implication that relative movements occurred between them during this interval. The Gondwana path is unfortunately less well-defined over this time interval (Torsvik et al., 2012) although within the limitations of the present data, the APW defined from Tarim and the Gondwana reference frame may not differ substantially (Fig. 10). The present record from Gondwana defines a large APW motion of ~60° for this supercontinent between 450 and 410 Ma which is broadly comparable to a contemporaneous motion identified in Laurentian data (Van der Voo, 1994; Piper, 2006) and may be attributable to True Polar Wander (TPW); it would then be recorded as a signature within the entire continental crust including Tarim. The rapid motion recognized in the Tarim Block was likely concentrated within a more limited time frame, apparently less than ~30 Ma of Late Ordovician-Wenlock times (Table 3), than the large motion defined by Torsvik et al. (2012) and other authors between ~450–410 Ma APW; a more precise estimate of the interval embraced by this motion will hinge on the timing of remanence acquisition in the Silurian Tataertage and Imogantau formations (Section 5.1.2).

In the context of the paleolatitude comparison between Tarim and contemporaneous early Paleozoic configurations of Laurentian and Gondwanan supercontinents (Fig. 11), we find that Tarim had reached low Northern Hemisphere paleolatitudes comparable to the North and South China blocks by mid-Silurian times (Fig. 11b). A significant change compared with our earlier analysis (e.g., Huang et al., 1999, 2000; Sun and Huang, 2009) is that we now place the North China, South China and Tarim blocks all at the western margin, rather than on both sides, of East Gondwana (Huang et al., 2018). There is a marked contrast in the paleogeographic settings of these blocks between mid-Ordovician and Wenlock times. During the earlier period Tarim was evidently sited marginally south of North China which in turn, then lay south of the South China Block (Fig. 11a). Note that Tarim and North China are sited specifically from the paleomagnetic data in this reconstruction (Huang et al., 1999, 2018 and this study) whilst the South China and Indochina blocks can be located from a wider range of evidence suggesting long proximity (750–380 Ma) between South China and Australia (Yang et al., 2004; Han et al., 2015) together with an early Paleozoic South China-Indochina connection (Cai and Zhang, 2009; Cocks and Torsvik, 2013). Fig. 11 implies substantial intra-early Paleozoic movements and whilst an origin of Tarim near the southern Eurasian margin in the vicinity of Siberia and Kazakhstania is permissible on paleomagnetic grounds alone, we strongly favor a position close to the North and South China blocks and near to the Indo-Arabian

Table 3
Summary of late Neoproterozoic-Paleozoic paleomagnetic poles from the Tarim block.

No.	Rock unit	Age (Ma)	Location		Lithology	Pole Position		A_{95} or dp/dm	Q	Reference
			°N	°E		°N	°E			
1	Tereeken formation	~635	41.5	87.8	Lst	27.6	140.4	8.8/11.1	6	Zhao et al. (2014)
2	Sugetbrak formation	~625	41.0	80.0	L + S	19.1	149.7	9.3	6	Zhan et al. (2007)
3	Zhamoketi formation	~615	41.5	87.8	L	−4.9	146.7	3.0/5.2	5	Zhao et al. (2014)
4	Baiyungang formation	485–478	41.3	83.4	Lst	−20.4	180.6	8.5/15.0	4	Fang et al. (1996)
5	Qiliang and Saergan Fm	467–453	40.6	78.9	Lst	−33.7	185.0	2.7/4.0	6	Huang et al. (this study)
6	Tataertage and Imogantau Fm	439–427	40.5	78.7	S	19.1	172.9	5.5	6	Huang et al. (this study)
7	Imogantau Fm	433–427	40.6	79.6	Siltst	12.1	158.4	4.1/7.2	6	Fang et al. (1996)
8	Tataertage, Imogantau, Kezirtage	439–383	40.6	79.0	Lst	16.5	165.2	2.9/5.3	5	Li et al. (1990)
9	Kezirtage Fm	427–383	40.5	78.8	S	9.8	152.7	3.6/5.5	4	Fang et al. (1996)
10	Qizilafu Fm	383–359	*		Siltst	13.1	156.5	4.7/8.0	4	Meng (1991)
11[a]	Heshilafu and Kelitage Fm	359–323	37.6	76.5	Lst	67.3	218.4	5.2/9.2	4	Fang et al. (2006)
12	Artashi of Shache, SW Tarim	359–299	15 levels		Lst	61.5	167.5	2.0/4.1	4	Meng (1991)
13	Duwa of Pishan, SW Tarim	359–299	15 levels		Lst	59.0	185.7	4.5/9.6	4	Meng (1991)
	Mean pole	329	2 poles			60.6	177.0	–		This study
14	Tagarqi and Azgan Fm	323–273	37.7	76.6	Lst	59.8	179.6	6.9/10.6	4	Fang et al. (2006) (recalculated)
15	Tagarqi, Pusige, and Duwa Fm	323–259	37.0	79.0	Lst + S	64.6	166.5	6.3	4	Gilder et al. (1996)
16	Kangkelin Fm (Longlinian)	299–273	40.0	79.0	Lst	46.6	170.2	4.3/6.1	4	Fang et al. (1991, 1996)
17	Kangkelin Fm (Longlinian)	299–273	40.5	78.8	Lst	41.2	160.0	8.0	4	Zhai et al. (1988) (recalculated)
18	Kangkelin Fm (Longlinian)	299–273	41.0	79.0	S	52.2	179.5	8.2/13.1	4	Bai et al. (1987)
	Mean pole	286	3 poles			46.9	169.2	13.2		This study
19	Qipan and Keziliqiman Fm (Longlingian to Chinhsian)	299–259	37.7	76.6	Lst	66.5	191.6	4.0/6.5	4	Fang et al. (2006) (recalculated)
20	260–290 Ma (P1-P2)	299–259	36.5	77.0	L + D	60.6	188.0	13.9/23.3	4	Li et al. (1995)
21	260–290 Ma (P1-P2)	299–259	39.8	78.8	D	65.6	181.2	3.9	5	Li et al. (1988)
	Mean pole	279	3 poles			64.3	186.9	5.9		This study
22[a]	Daliyuer Fm (late Chinhsian)	273–259	38.4	76.4	Silst + S	85.0	199.4	12.4	5	Gilder et al. (2008)
23	Kupukuziman Fm (P2, Artinskian to early Chinhsian)	273–259	40.6	79.5	L + S	53.9	178.5	2.6/4.1	5	Fang et al. (1991, 1996)
24	Kaipaizileike Fm (P2, late Chinhsian to Mokouan)	273–259	40.6	79.5	L + Siltst	61.7	185.2	5.5/8.5	5	Fang et al. (1991, 1996)
25	Shajingzi Fm (P2, early Tatarian)	273–259	40.4	79.4	S	55.8	179.2	4.5/7.0	5	Fang et al. (1991, 1996)
26	Biyouliti Gr. (Kalender and Baliqlig Fm., P2)	273–259	40.6	79.5	L + Lst	54.5	172.3	4.0	5	Sharps et al. (1989); Li et al. (1989)
27[a]	Baliqliq Fm. (P2, Artinskian to early Chinhsian)	273–259	40.5	78.8	Lst	65.1	165.5	4.1	5	Sharps et al. (1989); Li et al. (1989)
28	Kaipaizileike and Kupukuziman Fm (P2)	273–259	40.6	79.0	L + S	56.5	190.1	3.4/5.5	5	Bai et al. (1987)
29	Xiaotikanlik Fm (P2, Artinskian to early Chinhsian)	273–259	42.3	83.2	L	51.7	184.6	14.3/22.9	5	Fang et al. (1991)
30	Biyoulebaoguz Fm (P2, late Tatarian)	273–259	42.3	83.2	S	56.7	179.7	5.6/8.6	5	Fang et al. (1991)
31[a]	Tatarian (late Guadalupian, P2)	273–259	42.1	82.6	S + Siltst	73.2	191.0	7.2/10.3	5	McFadden et al. (1988)
	Mean pole	266	7 poles			55.9	181.3	3.4		This study

Footnotes: Q is the Quality Factor after Van der Voo (1990). Lithologies are: L = Lava, D = Dolerite, Lst = Limestone. S = Sandstone, Siltst = siltstone/mudstone/claystone. Precise location and lithology information unavailable for Meng (1991). The dp/dm are the radii of 95% confidence about the pole position in the colatitude direction and at right angle to it respectively. *Artashi of Shache, SW Tarim, 14 levels sampled, precise location unknown. Numerical ages (Ma) for Paleozoic paleomagnetic poles, constrained by geological epochs and stages, are referred to the International Chronostratigraphic Chart v2018/08 (http://www.stratigraphy.org/ICSchart/ChronostratChart2018-08.pdf), whilst numerical ages of the two Ordovician poles are according to more recent Ordovician integrative stratigraphy of China (Zhang et al., 2019).

[a] Paleomagnetic poles were not used in the construction of the apparent polar wander path for Tarim.

margin of Gondwana because the paleontological evidence shows that trilobite genera in Tarim, North China, South China and Qaidam–Qilian were initially related during the Ordovician within a single faunal province along the PAO margin (Zhou and Zhen, 2008; Zhou et al., 2009). Between Early Ordovician (Floian, ~478–470 Ma) and Late Ordovician (Katian, 453–445 Ma) times this uniform biogeographic province appears to have been dismembered to form two separate subprovinces embracing South China-Indochina-Tarim and North China-Sibumasu-southern Tibet respectively. Whilst these specific links are not all yet evident within the paleomagnetic constraints (Fig. 11), the regional context is clearly one involving large and differential dextral motions of blocks along the PAO margin of Gondwana (Huang et al., 2018). Other blocks such as Iran and Turkey are likely to have shared in this tectonic scenario (e.g., Burrett et al., 2014; Zhao et al., 2018).

By Silurian times the South China Block remained located in the vicinity of Australia (Han et al., 2015; Huang et al., 2018) and the broad proximity of Tarim with the other China blocks (Fig. 11b) is endorsed by the Silurian shelly faunas of China (Rong et al., 2003). These latter faunas identify Tarim, North China, South China, and Qaidam collectively as an assemblages moving northwards in unison as characterized by a common *Retziella* fauna and remained located within equatorial latitudes during the Silurian. A late early Paleozoic tectonic setting with latitudinal juxtaposing of the Tarim and North China blocks subjected to active margin tectonism with island arc accretion and magmatism, and accompanied by the opening of back-arc basins at the margin of the PAO has been reviewed in depth by Wilhem et al. (2012), Xiao et al. (2015, 2018) and Sengör et al. (2018) and is now complemented by the sequence of paleogeographic maps of Zhao et al. (2018) derived from both geological and paleomagnetic constraints.

A position of Tarim at ~26°N in the Northern Hemisphere in proximity to the Kazakhstan continent by Carboniferous times is recognized by the post-Silurian polar record (Bai et al., 1987; Fang et al., 1991). Full suturing between Tarim and Kazakhstania, and in turn with the southern margin of the Siberia Plate, seems likely to have been achieved by Permian times and was probably motivated by strong compression resulting from rapid northward movement of the Kunlun Terrane (Li et al., 1988, 1989; Sharps et al., 1989; Zhao et al., 1990; Fang et al., 1998; Yi et al., 2015). We note that the large intra-Carboniferous APW motion observed in the collective APWP of Gondwana (Torsvik et al., 2012) appears to replicate, at least in part, the path from Tarim (Table 3 and Fig. 10), although relative motions are also likely and are anticipated from the ongoing accretion and consolidation of the East Asian continental assemblage. Specific documentation of the early Paleozoic accretion-collision history of the Kazakhstan continent is

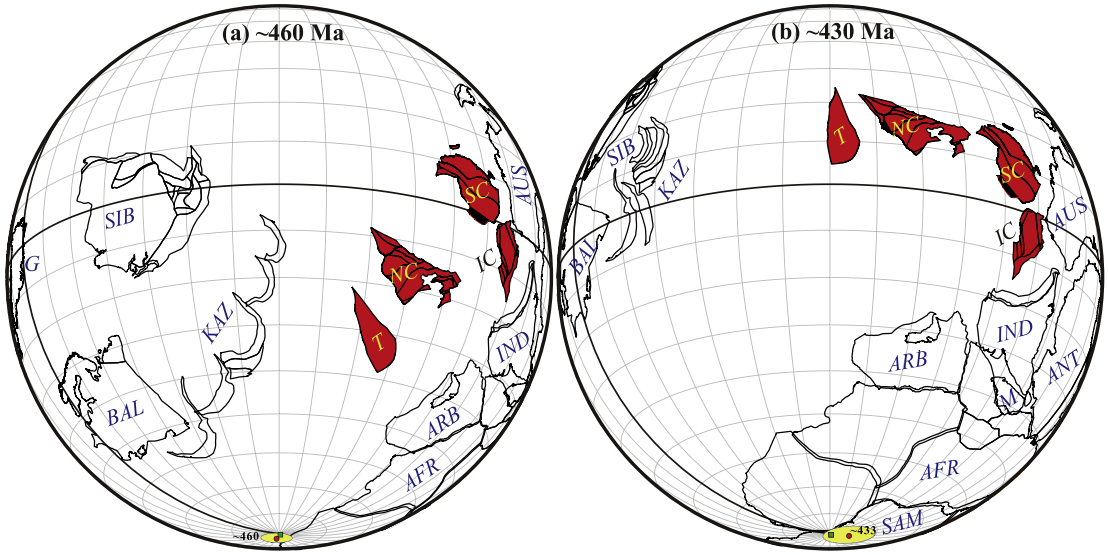


Fig. 11. Orthogonal stereographic projections showing paleogeographic reconstructions of the Tarim (T), North (NC) and South (SC) China, and Indochina (IC) blocks with respect to Gondwana at (a) ~460 Ma and (b) ~430 Ma shown in present geographic coordinates. The latitudinal positions of Tarim and North China are sited from the paleomagnetic data of Huang et al. (1999, 2000) and this study, whilst the South China and Indochina blocks are located using a postulated long term connection (750–380 Ma) between South China and Australia (Yang et al., 2004; Han et al., 2015) and a proposed early Paleozoic South China-Indochina connection (Cai and Zhang, 2009; Cocks and Torsvik, 2013). Greenland (G), Siberia (SIB), and Baltica (BAL) are shown following Torsvik et al. (2012) and the running mean ~460 Ma and ~430 Ma paleomagnetic poles for Gondwana in Australian coordinates (Torsvik et al., 2012) are used to rotate the reconstructions into present geographic coordinates, whilst the Kazakhstan continent is sketchily shown according to the conceptual ideas of Sengör et al. (1993). Euler rotation poles to Australia are: Tarim (T), ~430 Ma, 29.9°N, 118.5°E, Δ = 38.8°, ~460 Ma, 4.3°N, 122.2°E, Δ = 54.4°; North China (NC), ~430 Ma, 31.4°N, 86.7°, Δ = -79.4°; 460 Ma, 27.4°S, 275.9°E, Δ = 111.3°; South China (SC), 10.4°N, 140.0°E, Δ = 71.0° (Yang et al., 2004; Han et al., 2015); Indochina (IC): 2.6°S, 121.8°E, Δ = 101.5°. Other symbols are as for Fig. 10. The Euler rotated paleopoles (south pole) for both Gondwana (green squares) and Tarim (red circles with 95% confidence limits) are also shown in each reconstruction plot. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

covered by several recent reviews (e.g., Bazhenov et al., 2012; Wilhem et al., 2012; Xiao et al., 2015, 2018; Sengör et al., 2018), whilst aspects of the closure of the PAO and associated individual collisions of Tarim and North China blocks with Baltica and Siberia are discussed by Han and Zhao (2018) and Eizenhöfer and Zhao (2018).

6. Concluding remarks

The paleomagnetic results of this study, in conjunction with previously reported late Neoproterozoic and Paleozoic data from the Tarim Block lead to the following salient conclusions.

1. Together with our earlier study of this rock suite, the full Ordovician marine sediment collection from the Aksu-Kalpin-Bachu areas of northwestern Tarim records two distinctively different characteristic remanences. An A component is independent of rock formation age and lithology, yields a negative fold test result, and is interpreted as a Cenozoic overprint resulting from extensive tectonic deformation of this age in central Asia. A characteristic B component exists only in small amounts of relatively pure limestones but yields significant positive fold test results. It contrasts with Silurian and younger paleopoles from Tarim and is interpreted as a primary or quasi-primary component of Ordovician age.
2. Seventeen out of 21 Llandovery to Wenlock sites in fine-grained sandstones and mudstones yield a high-temperature characteristic remanence. Seven of these site-mean directions from the Aksu Sishichang section appear to be contaminated by the regional Permian igneous activity but 10 sites from the Kalpin Subhash section show no such influence and yield a paleopole supported by a positive fold test with a position contrasting with all younger poles from Tarim.
3. The new early Paleozoic paleomagnetic results indicate that Tarim was sited near the Arabian margin of East Gondwana at a ~32°S southerly paleolatitude at ~460 Ma. It then appears to have moved rapidly northwards by dextral motion along the Paleo-Asian margin so that by the Wenlock times (~430 Ma) it had become relocated around 9°N in northern hemisphere latitudes in common with the North and South China blocks in a setting close to the Australian margin of Gondwana. Subsequent suturing between Tarim and Kazakhstania and the southern margin of the Siberia Plate was achieved by Late Permian times and defined the closure of this sector of the Paleo-Asian Ocean.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.tecto.2019.02.010.

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